

Document 05 — Entity Model and Local Architecture

Backend Schema

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This project does not have a conventional web-app backend. Home Assistant is the local hub, and ESPHome devices communicate with it through the native API.

Architecture

```
Door-side ESPHome dial (ESP32-S3)
Bedside ESPHome sensor (ESP32-C6)
  ↓ WiFi LAN
Home Assistant
  ↓ local LAN integration
Tuya RGB bulbs
```

The ESPHome nodes are mostly stateless. They send user intent to Home Assistant and optionally display the current state.

I2C bus architecture

Each ESPHome node has its own independent I2C bus. Sensors are connected via STEMMA QT / Qwiic daisy-chain within each node.

Door-side node I2C bus

SENSOR	I2C ADDRESS	ESPHOME PLATFORM	ENTITY TYPE
TSL2591	0x29	tsl2591 (native)	Sensor (lux)
SHT45	0x44	sht4x (native)	Sensor (temperature, humidity)

Bedside node I2C bus

SENSOR	I2C ADDRESS	ESPHOME PLATFORM	ENTITY TYPE	V1 STATUS
APDS-9960	0x39	apds9960 (native)	Binary sensor (gestures), sensor (proximity)	v1 active
VL53L4CD	0x29	Needs verification	Sensor (distance in mm)	v1 deferred — Option B (2026-05-25); no v1 firmware references VL53L4CD

TSL2591 and VL53L4CD both use address 0x29 but are on different ESP32 nodes, so there is no conflict in the first-build architecture.

v1 scope note: All rows below that reference VL53L4CD as a source (or VL53L4CD-derived states such as `hand_present`, `hand_holding`, `nightlight_triggered`, `bedside_distance`, `bedside_hand_near`, `bedside_nightlight_hold`) describe the **planned v2 / future** schema. v1 does **not** materialize these entities — see `docs/vl53l4cd_support_verification.md`.

Home Assistant lighting model

Light group

Recommended group:

- `light.bedroom_group`

Members:

- Five Tuya RGB bulb entities.
- Actual entity IDs may differ after LocalTuya setup.

Reason for grouping:

- One command controls all bulbs.
- ESPHome YAML stays simple.
- Future bulbs can be added in Home Assistant without changing firmware.

ESPHome entities

Door-side dial (ESP32-S3)

Expected entities:

- Rotary position sensor.
- Rotary button sensor.
- Touch wake event.
- Backlight control.
- Optional LED ring light entity.
- `sensor.room_ambient_light` — TSL2591 lux reading. Used internally for adaptive backlight PWM and exposed to Home Assistant for dashboard/automations.
- `sensor.room_temperature` — SHT45 temperature in °C. Exposed to Home Assistant for dashboard display and future automations.
- `sensor.room_humidity` — SHT45 relative humidity in %. Exposed to Home Assistant for dashboard display and future automations.

Bedside gesture node (ESP32-C6)

Expected entities:

- Left gesture binary sensor. **(v1 active)**
- Right gesture binary sensor. **(v1 active)**
- Proximity sensor (APDS-9960). **(v1 active — diagnostic)**
- `sensor.bedside_distance` — VL53L4CD distance reading in mm. Used internally for hand-hold nightlight detection and exposed to Home Assistant for diagnostics. **(v1 deferred — VL53L4CD Option B, 2026-05-25; not materialized in v1 firmware.)**

Exact generated names depend on ESPHome naming and friendly-name settings.

Derived entities and logic states

These are not raw sensor readings. They are computed states derived from sensor data, used to drive UI behavior, automations, or firmware logic.

Door-side node derived states

DERIVED STATE	SOURCE	LOGIC	USED BY
<code>display_brightness_target</code>	TSL2591 lux	Lux-to-PWM mapping curve. Low lux (< 5) → backlight at 10%. High	

DERIVED STATE	SOURCE	LOGIC	USED BY
		lux (> 200) → backlight at 100%. Smooth interpolation between.	Backlight PWM output
<code>ambient_mode</code>	TSL2591 lux	<code>dark</code> if lux < 5, <code>dim</code> if 5–50, <code>bright</code> if > 50.	UI contrast/ accent adaptation, LED ring intensity
<code>display_awake</code>	Touch event, rotary event, or timer	<code>true</code> after any input. Reverts to <code>false</code> after idle timeout (default: 60 s).	Backlight on/off, LVGL page visibility
<code>idle_timer</code>	Last input timestamp	Counts seconds since last user interaction. Resets on any touch, rotation, or press.	Display sleep logic
<code>active_theme</code>	Theme Selector	Stores the currently active theme index (0–19). Persists across reboots.	UI rendering, LED ring color

Bedside node derived states

DERIVED STATE	SOURCE	LOGIC	USED BY	V1 STATUS
<code>hand_present</code>	VL53L4CD distance	<code>true</code> if distance < 300 mm for > 100 ms. <code>false</code> otherwise.	APDS-9960 wake trigger	v1 deferred (VL53L4CD Option B)
<code>hand_holding</code>	VL53L4CD distance + timer	<code>true</code> if distance is 50–100 mm continuously for > 1.5 s.	Nightlight scene trigger	v1 deferred (VL53L4CD Option B)
<code>gesture_ready</code>	<code>hand_present</code> state	<code>true</code> when <code>hand_present</code> is <code>true</code> and APDS-9960 is powered up and ready to detect.	Gesture detection gate	v1 deferred (depends on VL53L4CD; v1 APDS gestures run unconditionally)
<code>nightlight_triggered</code>	<code>hand_holding</code> transition	One-shot <code>true</code> on rising edge of <code>hand_holding</code> .	HA service call:	v1 deferred (VL53L4CD Option B)

DERIVED STATE	SOURCE	LOGIC	USED BY	V1 STATUS
		Resets after action is sent. Cooldown of 5 s prevents re-trigger.	nightlight scene	
<code>gesture_cooldown</code>	Last gesture timestamp	Blocks new gesture detection for 1 s after a successful gesture to prevent double-fires.	Gesture debounce	v1 active

Home Assistant derived entities (optional, created via template sensors)

ENTITY	SOURCE	LOGIC	PURPOSE
<code>binary_sensor.bedroom_occupied</code>	VL53L4CD activity + gesture history	<code>true</code> if any bedside interaction in the last 30 minutes.	Future automations (auto-off after leaving)
<code>sensor.bedroom_comfort_index</code>	SHT45 temperature + humidity	Simple comfort formula (e.g., heat index or PMV approximation).	Dashboard display, future HVAC hints

These Home Assistant template entities are optional and not required for the first build. They are documented here for future expansion.

Example Home Assistant/template entity names

The following table lists the expected entity IDs as they will appear in Home Assistant after ESPHome integration. Actual names depend on ESPHome `name:` and `friendly_name:` settings, but these are the recommended defaults.

Door-side node entities (ESPHome device name: `veladial_doorside`)

ENTITY ID	TYPE	SOURCE	UPDATE RATE	NOTES
<code>sensor.veladial_doorside_ambient_light</code>	sensor	TSL2591	2 s	Lux value, 0–88000 range
<code>sensor.veladial_doorside_temperature</code>	sensor	SHT45	30 s	

ENTITY ID	TYPE	SOURCE	UPDATE RATE	NOTES
				°C, ±0.1° accuracy
<code>sensor.veladial_doorside_humidity</code>	sensor	SHT45	30 s	%, ±1% accuracy
<code>sensor.veladial_doorside_backlight_level</code>	sensor	Internal	On change	Current backlight PWM %
<code>binary_sensor.veladial_doorside_display_awake</code>	binary_sensor	Internal	On change	Whether display is active
<code>sensor.veladial_doorside_wifi_signal</code>	sensor	WiFi	60 s	dBm signal strength
<code>text_sensor.veladial_doorside_active_theme</code>	text_sensor	Internal	On change	Name of the currently active UI theme

Bedside node entities (ESPHome device name: `veladial_bedside`)

ENTITY ID	TYPE	SOURCE	UPDATE RATE	NOTES	V1 STATUS
<code>binary_sensor.veladial_bedside_gesture_left</code>	binary_sensor	APDS-9960	On event	Momentary pulse on left swipe	v1 active
<code>binary_sensor.veladial_bedside_gesture_right</code>	binary_sensor	APDS-9960	On event	Momentary pulse on right swipe	v1 active
<code>sensor.veladial_bedside_proximity</code>	sensor	APDS-9960	200 ms	Raw proximity 0-255	v1 active (diagnost
<code>sensor.veladial_bedside_distance</code>	sensor	VL53L4CD	200 ms		

ENTITY ID	TYPE	SOURCE	UPDATE RATE	NOTES	V1 STATUS
				Distance in mm, 0–1300 range	v1 deferred (VL53L4CD Option B)
<code>binary_sensor.veladial_bedside_hand_present</code>	binary_sensor	VL53L4CD (derived)	On change	Hand detected within 300 mm	v1 deferred (VL53L4CD Option B)
<code>binary_sensor.veladial_bedside_hand_holding</code>	binary_sensor	VL53L4CD (derived)	On change	Hand held at 50–100 mm for > 1.5 s	v1 deferred (VL53L4CD Option B)
<code>sensor.veladial_bedside_wifi_signal</code>	sensor	WiFi	60 s	dBm signal strength	v1 active

Home Assistant template entities (optional, configured in HA not ESPHome)

ENTITY ID	TYPE	TEMPLATE LOGIC	REQUIRED FOR FIRST BUILD
<code>binary_sensor.bedroom_occupied</code>	binary_sensor	<code>true</code> if any bedside interaction in last 30 min	No
<code>sensor.bedroom_comfort_index</code>	sensor	Heat index from temperature + humidity	No

Actions

Initial Home Assistant actions:

- Turn the group off for a left gesture. **(v1 active)**
- Turn the group on for a right gesture. **(v1 active)**
- Turn the group on at low brightness and warm color for nightlight (triggered by VL53L4CD hand-hold at 5–10 cm for 1.5 s). **(v1 deferred — VL53L4CD Option B, 2026-05-25.)**
- Set brightness from the rotary control. **(v1 active)**

Sensor-driven local behaviors (internal to ESPHome, not HA actions)

- TSL2591 lux reading → door-side backlight PWM adjustment (adaptive brightness). **(v1 active.)**

- VL53L4CD distance → APDS-9960 wake trigger (sensor fusion on bedside node). **(v1 deferred — VL53L4CD Option B; sensor fusion is independently a v2 / future enhancement.)**

These behaviors run inside the ESPHome firmware and do not require Home Assistant round-trips.

Sensitive data

Do not commit live WiFi credentials, API credentials, Home Assistant tokens, Tuya local keys, or generated ESPHome build output. Keep private values in the local ESPHome/Home Assistant configuration.

Configuration values to keep easy to change

- Light group entity ID.
- Gesture threshold and cooldown.
- Screen dim timeout.
- Brightness step size.
- Color presets.
- Rotary encoder direction.
- Touch transform settings.
- TSL2591 lux-to-backlight mapping curve.
- VL53L4CD hand-hold distance threshold (default: 5–10 cm).
- VL53L4CD hand-hold duration threshold (default: 1.5 s).
- SHT45 update interval.
- Active theme index (0-19, persists across reboots).
- Theme selector timeout (default: 60 s).
- Long-press duration to enter Theme Selector (default: 1.5 s).
- Per-theme LED ring RGB color values.

Generic example entity aliases

GENERIC EXAMPLE ENTITY	MAPS TO / DERIVED FROM	PURPOSE
<code>binary_sensor.bedside_hand_near</code>	<code>binary_sensor.veladial_bedside_hand_present</code> / VL53L4CD distance	Human-friendly generic alias for hand-near detection

GENERIC EXAMPLE ENTITY	MAPS TO / DERIVED FROM	PURPOSE
<code>binary_sensor.bedside_nightlight_hold</code>	<code>binary_sensor.veladial_bedside_hand_holding</code> / VL53L4CD distance + timer	Human-friendly generic alias for deliberate nightlight hold
<code>number.display_auto_brightness_level</code>	<code>display_brightness_target</code> / TSL2591 lux mapping	Optional mapped display brightness target
<code>sensor.last_sensor_update</code>	Sensor timestamp diagnostics	Optional helper for stale sensor detection
<code>binary_sensor.sensor_health_ok</code>	Sensor availability checks	Optional helper for sensor health diagnostics

These are example aliases/helper names only. Final production entity IDs depend on ESPHome names, Home Assistant helper setup, and firmware implementation.

Sensor and entity behavior mapping

INPUT/ENTITY	CONDITION	ACTION	V1 STATUS
TSL2591 lux	low ambient light	dim display/backlight	v1 active
TSL2591 lux	bright room	increase display readability	v1 active
VL53L4CD distance	hand steady 5–10 cm for about 1.5 s	trigger nightlight mode	v1 deferred (Option B)
APDS gesture left	valid gesture + cooldown clear	turn bedroom group off	v1 active

INPUT/ENTITY	CONDITION	ACTION	V1 STATUS
APDS gesture right	valid gesture + cooldown clear	turn bedroom group on	v1 active
SHT45 temp/ humidity	available	expose diagnostics only	v1 active (secondary)

Security and privacy model

- No cloud backend is required.
- Sensor data stays inside Home Assistant/local network unless the owner later adds cloud integrations.
- No camera is used.
- No microphone is used.
- No audio recording is used.
- No voice dependency is used.
- Environmental data is diagnostic/comfort data only.

Scope guard

- Temperature/humidity remains secondary.
- Environmental data must not clutter the main Power/Brightness/Presets UI.
- Environmental data is not part of the core lighting control loop.
- No HVAC/comfort automation is required in the first build.
- The main product remains a silent local-first lighting controller, not a general environmental monitor.

Open implementation questions

- Confirm final ESPHome entity names after firmware compile.
- Verify VL53L4CD ESPHome support path.
- Decide exact lux-to-backlight mapping curve after real room testing.
- Decide exact distance threshold and hold time after bedside testing.
- Decide whether hand-near/nightlight-hold states live fully in ESPHome or are exposed to Home Assistant as helpers.
- Decide how often to publish SHT45 values to Home Assistant.
- Decide whether sensor health diagnostics are implemented in ESPHome or Home Assistant.